

Look Ahead Before You Distill: Future Trajectory Validation of Teacher Guidance for Agentic On-Policy Distillation

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Abstract

On-policy distillation (OPD) provides teacher supervision on states visited by the student, reducing the distribution gap between training and inference. However, in multi-turn agentic tasks, student deviations may accumulate over time, gradually moving the trajectory away from states where teacher guidance remains effective. Our quantitative analysis further shows that high-disagreement states offer promising opportunities for teacher guidance, but determining whether such guidance is beneficial requires examining its effect on subsequent student trajectories. We propose **FutureBridge-OPD (FTB)**, which executes a short teacher bridge at a high disagreement state and uses the resulting student continuation to assess whether the bridge increases the density of positive distillation signals relative to the teacher. On ALFWorld, WebShop, and ScienceWorld, under the main Qwen3-32B teacher to Qwen3-1.7B student setting, FTB outperforms vanilla OPD and TCOD by an average of **16.6** and **7.6** points, respectively, and remains effective across student scales and teacher settings. Our code is publicly available at <https://github.com/ChenChiShui/FutureBridge-OPD>.

1 Introduction

On-policy distillation (OPD) has recently emerged as an effective approach for transferring the capabilities of large language models to smaller models Agarwal et al. (2024). By providing teacher supervision on states visited by the student itself, OPD alleviates the distribution mismatch between offline distillation and actual inference Ross et al. (2011); Bengio et al. (2015). However, in multi-turn agentic tasks, early deviations made by the student alter the subsequent state distribution and accumulate along the trajectory, gradually weakening the effectiveness of later teacher supervision Zhong et al. (2026); Liao et al. (2026).

Existing agentic OPD methods mainly tackle this issue in two ways. Methods such as TCOD and Guided-OPD regulate student takeover through teacher prefixes, curriculum schedules, or adaptive rollout depths Wang et al. (2026a); Li et al. (2026a); Zhou et al. (2026b). Another line of work filters or reweights distillation signals using KL divergence, model confidence, or environment feedback Xu et al. (2026b); Zheng et al. (2026); Zhou et al. (2026a). The former reshapes the student-visited state distribution, whereas the latter estimates supervision value from local information at the current state. However, both largely overlook one question: *Can teacher guidance steer the student’s future trajectory toward regions with higher teacher preference?*

To study this question, we analyze 1,000 failed trajectories from ALFWorld, WebShop, and ScienceWorld Shridhar et al. (2021); Yao et al. (2022); Wang et al. (2022). We replay the history prefix, replace the student action at positions with large teacher–student disagreement, and then let the same student policy complete the remaining interaction. Further details are given in Appendix B. This analysis reveals two useful properties of teacher guidance in multi-turn agentic OPD:

(I) Key Action Replacement Improves Outcomes. As illustrated in the second panel of Figure 1, replacing the student action with the teacher action at positions with large teacher–student KL divergence improves environment rewards in approximately **31%** of the trajectories, compared with only **6%** when actions are replaced at random positions. However, even among these high-KL positions, approximately **19%** of the interventions still degrade trajectory performance. This indicates that teacher–student KL divergence is useful for identifying candidate intervention positions, but insufficient for determining whether the corresponding teacher guidance is beneficial.

(II) Teacher Guidance Redirects Students Toward Teacher-Preferred Regions. We further examine how teacher guidance affects the continuation autonomously

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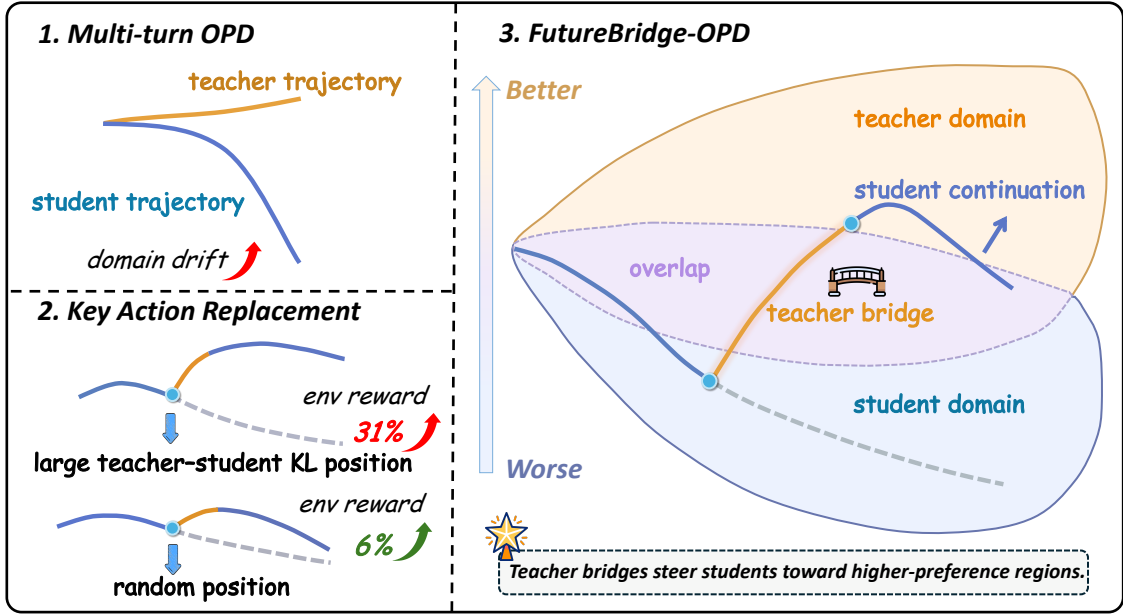


Figure 1: Motivation and overview of FutureBridge-OPD. (1) Student deviations accumulate over multi-turn interactions. (2) High KL positions offer promising intervention opportunities. (3) The induced student continuation validates the teacher bridge, which redirects the student toward teacher preferred regions.

generated by the student. After adopting the teacher action, the proportion of teacher preferred tokens in the subsequent student trajectory increases by approximately **10.8%** in relative terms, conceptually illustrated in the third panel of Figure 1 as the student continuation shifting toward the teacher preferred region. This result suggests that the effect of appropriate local guidance is not confined to the current decision, but persists across subsequent interactions. Therefore, assessing the value of candidate teacher guidance also requires looking ahead to the future student trajectory that it induces.

We thus propose **FutureBridge-OPD (FTB)**, a two-stage framework that locates candidate interventions through teacher-student disagreement and validates teacher guidance through the induced future student trajectory. FTB first identifies a candidate position and executes a *teacher bridge* by replacing the student decision with a teacher action. The same student policy then generates a short continuation from the bridged state, and the resulting future trajectory determines whether the guidance redirects subsequent student behavior toward regions with higher teacher preference and should be retained for distillation.

Our main contributions are summarized as follows:

- **Systematic Analysis of Teacher Guidance in Agentic OPD.** We analyze the interplay among student rollout depth, teacher-student disagreement, and future distillation signals in multi-turn agentic OPD. We show that high-disagreement interventions more often improve outcomes and steer future student trajectories toward denser positive distillation signals.

- **A Framework for Future Validation of Teacher Guidance.** We propose FutureBridge-OPD, which executes a teacher bridge at a candidate decision and validates it through the induced student continuation before distillation.
- **Consistent Gains across Tasks and Models.** Across ALFWorld, WebShop, and ScienceWorld, FutureBridge-OPD improves the average success rate by **16.6** and **7.6** points over vanilla OPD Agarwal et al. (2024) and TCOB-B2F Wang et al. (2026a), respectively, and remains effective across student scales and teacher settings.

2 Related Work

On-Policy Distillation for Agents. Knowledge distillation transfers the predictive distribution of a large teacher to a smaller student, enabling model compression and capability transfer Hinton et al. (2015). Distillation at the sequence level extends this principle to autoregressive sequence generation Kim and Rush (2016), while MiniLLM further investigates reverse KL objectives tailored to large language model generation Gu et al. (2024). On-policy distillation (OPD) provides teacher supervision on data generated and states visited by the student, thereby reducing the distribution mismatch between offline distillation and actual inference Agarwal et al. (2024).

Recent studies further examine the mechanisms, applicable conditions, and potential failure modes of OPD under long horizon generation Li et al. (2026b); Fu et al. (2026); Yang et al. (2026); Liu et al. (2026). For multi-turn agent tasks, TCOB uses a temporal curriculum to control how

deeply the student takes over the trajectory Wang et al. (2026a), while Guided-OPD mixes teacher and student turns according to a curriculum schedule Li et al. (2026a). Other studies improve trajectory construction and teacher guidance in multi-turn OPD through prefix replay, short-to-long rollouts, adaptive rollout depths, adjustment at the turn level, or environment feedback Liao et al. (2026); Zhang et al. (2026b); Zhou et al. (2026b); Tan et al. (2026); Zhou et al. (2026a).

Selective Supervision and Reweighting in OPD. Another line of work focuses on selecting or reweighting distillation supervision across different tokens, turns, or reasoning steps. Existing methods use signals such as generation position, teacher entropy, student uncertainty, causal lookahead, teacher-student divergence, trajectory outcomes, or model perplexity to identify high value supervision or adjust its distillation strength Zhang et al. (2026a); Jin et al. (2026); Ke et al. (2026); Xu et al. (2026b); Zheng et al. (2026); Zhong et al. (2026); Xing et al. (2026). These methods generally estimate supervision value primarily from the current response, local model distributions, or outcomes of trajectories already generated by the student itself.

Future Information for Supervision and Credit Assignment. In language model reasoning and policy optimization, several studies use future continuations to estimate the downstream effects of local decisions. Existing approaches reshape advantages at the token level using future KL divergence, perform detailed credit assignment through counterfactual perturbations or gradient approximations, resample continuations from intermediate reasoning states, or combine local uncertainty with future continuation signals for branching and guidance Ma et al. (2026); Li et al. (2026c); Samanta et al. (2026); Wang et al. (2026b); Jiang and Ferraro (2026). These studies demonstrate the value of future information for evaluating local decisions.

3 Method

3.1 Preliminary: On-Policy Distillation

We consider a finite horizon, multi-turn interaction between an agent and an environment. At turn t , the agent receives an observation o_t , generates a response a_t , and executes it as a single environment action. The environment then returns the next observation o_{t+1} . We denote the interaction history by $h_t = (o_1, a_1, \dots, o_{t-1}, a_{t-1}, o_t)$ and a complete trajectory by $\tau = (o_1, a_1, \dots, o_T, a_T, o_{T+1})$.

On-policy distillation (OPD) aligns a student policy π_θ with a teacher policy π_ϕ on states visited by the student, thereby reducing the mismatch between training and inference as follows:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{\tau \sim p_{\pi_\theta}} \left[\sum_{t=1}^T D_{\text{KL}}(\pi_\theta(\cdot | h_t) \| \pi_\phi(\cdot | h_t)) \right]. \quad (1)$$

For multi-turn tasks, we adopt the Back-to-Forward curriculum (B2F) from TCOd. Given a successful trajectory of length L , B2F replays its first $L - k$ turns and lets the student complete the remaining interaction, with k gradually increasing during training. Following TCOd’s public implementation, trajectories are collected using a frozen student policy $\pi_{\bar{\theta}}$, and Eq. (1) is optimized with a sampled token estimator during training.

For a student response $a_t = (x_{t,1}, \dots, x_{t,M_t})$, let $c_{t,i} = (h_t, x_{t,<i})$ be the context of token $x_{t,i}$. We denote the raw sampled distillation advantage by

$$A_{t,i} = \beta [\log \pi_\phi(x_{t,i} | c_{t,i}) - \log \pi_{\bar{\theta}}(x_{t,i} | c_{t,i})], \quad (2)$$

where $\beta > 0$ is the distillation coefficient. A positive value indicates that the teacher assigns the sampled token higher likelihood than the frozen student. We explicitly call $A_{t,i}$ *raw* here to distinguish it from clipped or normalized variants used during optimization.

3.2 FutureBridge-OPD

FTB uses a local sampled signal to propose a candidate teacher intervention and paired future continuations to validate its downstream effect, as illustrated in Figure 2.

Candidate Localization. Given a B2F student rollout τ , we score the response actually executed at each turn controlled by the student:

$$\hat{d}_t = \frac{1}{M_t} \sum_{i=1}^{M_t} \log \frac{\pi_{\bar{\theta}}(x_{t,i} | c_{t,i})}{\pi_\phi(x_{t,i} | c_{t,i})}. \quad (3)$$

Let $\bar{A}_t = M_t^{-1} \sum_{i=1}^{M_t} A_{t,i}$. By autoregressive factorization,

$$\hat{d}_t = \frac{1}{M_t} \log \frac{\pi_{\bar{\theta}}(a_t | h_t)}{\pi_\phi(a_t | h_t)} = -\frac{1}{\beta} \bar{A}_t.$$

We therefore select

$$t^* = \arg \max_{t \in \mathcal{T}_S(\tau)} \hat{d}_t = \arg \min_{t \in \mathcal{T}_S(\tau)} \bar{A}_t, \quad (4)$$

where $\mathcal{T}_S(\tau)$ contains the turns generated by the student after the replayed teacher prefix, excluding the final turn.

Thus, FTB selects the realized student response with the lowest average distillation advantage relative to the teacher. Here, $\hat{d}_t$ directly evaluates the response that induced the observed environment transition. It is a directional sampled score and is used only to localize a potentially correctable turn; whether the teacher replacement is beneficial is determined by the future validation below. At most one bridge is attempted per trajectory, bounding validation cost and isolating a single intervention.

Teacher Bridge and Paired Continuations. Let $a_{t^*}^{\text{stu}}$ denote the original student response. Conditioned on the same history h_{t^*} , the teacher generates a bridge response $a_{t^*}^{\text{br}}$ using the student’s decoding configuration. We restore the environment state at t^* , execute $a_{t^*}^{\text{br}}$ in place of $a_{t^*}^{\text{stu}}$, and return control to the frozen student $\pi_{\bar{\theta}}$ for an H -turn continuation ξ^{br} . The corresponding continuation following the original student response is denoted by ξ^{base} .

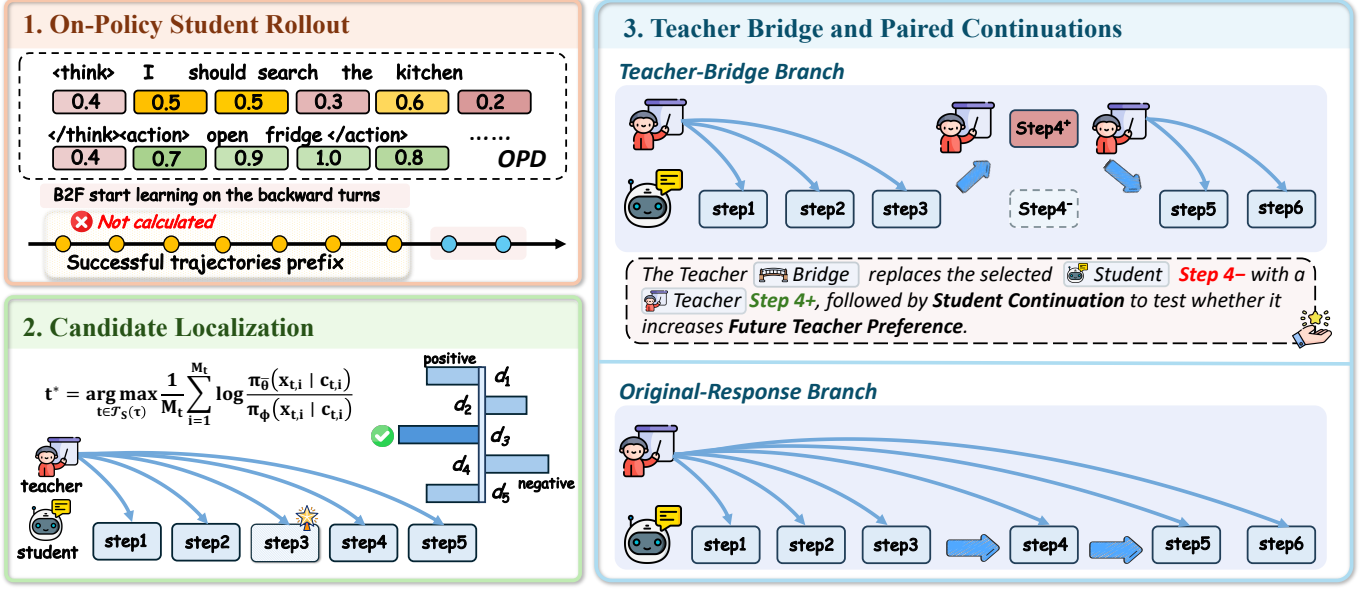


Figure 2: Overview of FutureBridge-OPD: (1) on-policy student rollouts provide distillation signals; (2) maximum teacher–student disagreement identifies a candidate bridge turn; (3) paired continuations validate the teacher bridge.

Both continuations start from the same history before intervention and, after the alternative actions, are generated by the same frozen student. Their difference therefore reflects the downstream effect of replacing the realized student response with the teacher bridge.

Future Validation Gate. For any sequence generated by the student $y = (y_1, \dots, y_{|y|})$, we define

$$\rho(y) = \frac{1}{|y|} \sum_{i=1}^{|y|} \mathbf{1}[A_i(y) > 0], \quad (5)$$

where $A_i(y)$ denotes Eq. (2) evaluated on token y_i and its context. Equivalently, $\rho(y)$ is the fraction of tokens to which the teacher assigns higher likelihood than the frozen student. It measures the teacher preferred token ratio along the generated student continuation. This ratio is normalized by the total number of generated tokens, facilitating comparisons between continuations of different lengths and preventing a small number of large log probability differences from dominating the aggregate statistic.

The bridge acceptance indicator is

$$g(\tau) = \mathbf{1}[\rho(\xi^{\text{br}}) > \rho(\xi^{\text{base}})]. \quad (6)$$

A bridge is retained only when its induced student continuation has a higher teacher preferred token ratio than the corresponding original continuation.

The paired continuations are used only as validation evidence. For an accepted bridge, only $a_{t^*}^{\text{br}}$ is added to the B2F experience buffer. Beyond the successful reference trajectories inherited from B2F, this procedure uses no additional environment reward or task success label.

3.3 Future Teacher Preference

Let $\xi_H(h, a; \pi_{\bar{\theta}})$ denote the continuation obtained by executing action a under history h and subsequently rolling

out the frozen student for H turns. We define an auxiliary future teacher preference value by

$$U_{\phi, \bar{\theta}}^H(h, a) = \mathbb{E}[\rho(\xi_H(h, a; \pi_{\bar{\theta}}))]. \quad (7)$$

It measures the expected teacher preferred token ratio in the student continuation induced by a .

For the teacher bridge and the original student response, define the future teacher preference gain

$$\Delta_{\phi, \bar{\theta}}^H = U_{\phi, \bar{\theta}}^H(h_{t^*}, a^{\text{br}}) - U_{\phi, \bar{\theta}}^H(h_{t^*}, a^{\text{stu}}), \quad (8)$$

with the paired sample estimate

$$\hat{\Delta}_{\phi, \bar{\theta}}^H = \rho(\xi^{\text{br}}) - \rho(\xi^{\text{base}}). \quad (9)$$

The gate retains bridges satisfying $\hat{\Delta}_{\phi, \bar{\theta}}^H > 0$, i.e., bridges with a positive sampled future teacher preference gain.

Candidate localization and future validation use complementary aggregations of the same sampled OPD signal. The signed average advantage localizes a realized response receiving weak teacher relative support, while the teacher preferred token ratio characterizes the student continuation induced by replacing that response.

Accepted teacher bridges are optimized with

$$\mathcal{L}_{\text{bridge}}(\theta) = -\mathbb{E} \left[\frac{g(\tau)}{M_{\text{br}}} \sum_{i=1}^{M_{\text{br}}} \text{sg}[A_i^{\text{br}}] \log \pi_{\theta}(x_i^{\text{br}} | c_i^{\text{br}}) \right]. \quad (10)$$

At $\theta = \bar{\theta}$, treating the accepted bridge data and gate as fixed, its gradient matches that of a sampled squared teacher–student log probability discrepancy objective. The formal statement and proof are provided in Appendix D.

Method	ALFWorld	WebShop		ScienceWorld	
	SR $\uparrow$	Score $\uparrow$	SR $\uparrow$	Score $\uparrow$	SR $\uparrow$
Qwen3-32B teacher $\rightarrow$ Qwen3-1.7B student					
Student (zero-shot)	7.5	39.0	31.0	6.4	2.0
Teacher (zero-shot)	50.0	52.8	51.0	28.8	12.0
OPD	21.5 $\pm$ 3.0	32.6 $\pm$ 2.2	31.7 $\pm$ 2.9	22.2 $\pm$ 1.3	2.0 $\pm$ 1.5
TCOD-F2B	30.7 $\pm$ 3.2 $\uparrow$ 9.2	39.6 $\pm$ 1.7 $\uparrow$ 7.0	38.3 $\pm$ 1.2 $\uparrow$ 6.6	26.1 $\pm$ 0.0$\uparrow$ 3.9	5.8 $\pm$ 0.8 $\uparrow$ 3.8
TCOD-B2F	35.0 $\pm$ 3.3$\uparrow$ 13.5	41.8 $\pm$ 3.7 $\uparrow$ 9.2	40.3 $\pm$ 3.3 $\uparrow$ 8.6	22.3 $\pm$ 0.4 $\uparrow$ 0.1	6.7 $\pm$ 0.6$\uparrow$ 4.7
Guided-OPD	33.1 $\pm$ 5.4 $\uparrow$ 11.6	44.6 $\pm$ 11.7 $\uparrow$ 12.0	41.7 $\pm$ 10.6 $\uparrow$ 10.0	23.5 $\pm$ 2.3 $\uparrow$ 1.3	4.7 $\pm$ 0.6 $\uparrow$ 2.7
FTB w/o Bridge Exec.	31.4 $\pm$ 3.2 $\uparrow$ 9.9	58.4 $\pm$ 3.4$\uparrow$ 25.8	55.3 $\pm$ 3.7$\uparrow$ 23.6	18.7 $\pm$ 3.3 $\downarrow$ 3.5	3.7 $\pm$ 1.2 $\uparrow$ 1.7
FTB (Ours)	40.4 $\pm$ 3.4$\uparrow$ 18.9	58.1 $\pm$ 1.5$\uparrow$ 25.5	57.7 $\pm$ 1.7$\uparrow$ 26.0	26.4 $\pm$ 0.4$\uparrow$ 4.2	6.8 $\pm$ 0.6$\uparrow$ 4.8
Qwen3-32B teacher $\rightarrow$ Qwen3-4B student					
Student (zero-shot)	20.9	32.9	29.0	23.1	8.5
Teacher (zero-shot)	50.0	52.8	51.0	28.8	12.0
OPD	42.3 $\pm$ 4.9	43.2 $\pm$ 1.7	40.3 $\pm$ 2.6	29.0 $\pm$ 0.3	7.3 $\pm$ 1.0
TCOD-F2B	47.5 $\pm$ 4.6 $\uparrow$ 5.2	49.7 $\pm$ 7.9 $\uparrow$ 6.5	46.7 $\pm$ 6.8 $\uparrow$ 6.4	23.9 $\pm$ 0.4 $\downarrow$ 5.1	11.3 $\pm$ 2.5 $\uparrow$ 4.0
TCOD-B2F	39.8 $\pm$ 4.8 $\downarrow$ 2.5	50.9 $\pm$ 1.2$\uparrow$ 7.7	47.3 $\pm$ 2.6 $\uparrow$ 7.0	24.5 $\pm$ 2.1 $\downarrow$ 4.5	9.5 $\pm$ 0.4 $\uparrow$ 2.2
Guided-OPD	47.0 $\pm$ 9.3 $\uparrow$ 4.7	47.7 $\pm$ 5.8 $\uparrow$ 4.5	47.0 $\pm$ 5.9 $\uparrow$ 6.7	28.3 $\pm$ 1.7 $\downarrow$ 0.7	11.3 $\pm$ 2.0 $\uparrow$ 4.0
FTB w/o Bridge Exec.	48.5 $\pm$ 9.2$\uparrow$ 6.2	50.7 $\pm$ 0.3$\uparrow$ 7.5	48.3 $\pm$ 1.2$\uparrow$ 8.0	24.6 $\pm$ 3.7 $\downarrow$ 4.4	12.5 $\pm$ 3.3$\uparrow$ 5.2
FTB (Ours)	48.3 $\pm$ 1.9$\uparrow$ 6.0	50.9 $\pm$ 5.9$\uparrow$ 7.7	50.0 $\pm$ 5.7$\uparrow$ 9.7	28.4 $\pm$ 2.9$\downarrow$ 0.6	14.0 $\pm$ 0.4$\uparrow$ 6.7

Table 1: Main results with Qwen3-32B as the teacher, reported as mean $\pm$ standard deviation over three seeds; annotations show absolute changes over OPD, while **blue** and **red** backgrounds denote the best and second-best trained methods, respectively.

4 Experiments

4.1 Experimental Setup

Benchmarks and Models. We conduct experiments on three multi-turn agent benchmarks: ALFWorld Shridhar et al. (2021), WebShop Yao et al. (2022), and ScienceWorld Wang et al. (2022). We follow TCOD Wang et al. (2026a) in using the same data splits, environment configurations, and prompt templates. The maximum numbers of environment interaction steps are set to 30, 15, and 30 for ALFWorld, WebShop, and ScienceWorld, respectively. Our main experiments use Qwen3-32B as the teacher and we further evaluate Qwen3-4B as the student and a Qwen3-8B teacher trained with reinforcement learning Yang et al. (2025); Shao et al. (2024); Feng et al. (2025). Further details of the teacher training are provided in Appendix C.3. We report success rate for ALFWorld, and both task score and success rate for WebShop and ScienceWorld.

Training Setup. All methods use the same training budget of 200 optimization steps. FTB is built upon the B2F curriculum introduced by TCOD Wang et al. (2026a), where the teacher prefix is shortened by one environment step every five training steps, allowing the student to progressively take control from earlier states. Further experimental details are provided in Appendix C. **Baselines.** We compare with several agentic on-policy distillation methods. Vanilla OPD directly performs teacher-student distillation on states visited by the student Agarwal et al. (2024). TCOD-F2B starts from the early part of the trajectory and progressively increases the student rollout depth, while TCOD-B2F initializes the student

from intermediate states using prefixes of successful reference trajectories and gradually shortens these prefixes during training Wang et al. (2026a). Guided-OPD mixes teacher and student turns through a curriculum schedule, progressively increasing the student’s control over the full trajectory Li et al. (2026a). All baselines are implemented following the core settings of their original papers.

4.2 Main Results

Results with the Qwen3-32B Teacher. As shown in Table 1, with the Qwen3-1.7B student, FTB improves the average success rate over OPD and TCOD-B2F by 16.6 and 7.6 percentage points, respectively. When scaling the student to Qwen3-4B, FTB still yields improvements of 7.5 and 5.2 percentage points, demonstrating consistent gains across student scales. Notably, on WebShop, the 1.7B student trained with FTB outperforms both the 4B student and the zero-shot 32B teacher. Trajectory analysis shows that the 1.7B student follows simpler and more direct paths, averaging 7.67 steps with a 32% maximum step rate, while the 4B student averages 11.37 steps with a 51% maximum step rate and more often becomes trapped in redundant interactions. FTB trains the student on validated local guidance and supports the development of a task specific policy with more efficient execution, helping the student surpass the zero-shot teacher.

Results with a Teacher Trained with RL. As shown in Table 2, when using the Qwen3-8B teacher trained with RL, FTB achieves the best performance on all three reported metrics. Its average success rate across ALFWorld and WebShop exceeds OPD and TCOD-B2F by 4.5 and 4.3 percentage points, respectively, confirming its effectiveness under different teacher configurations considered here.

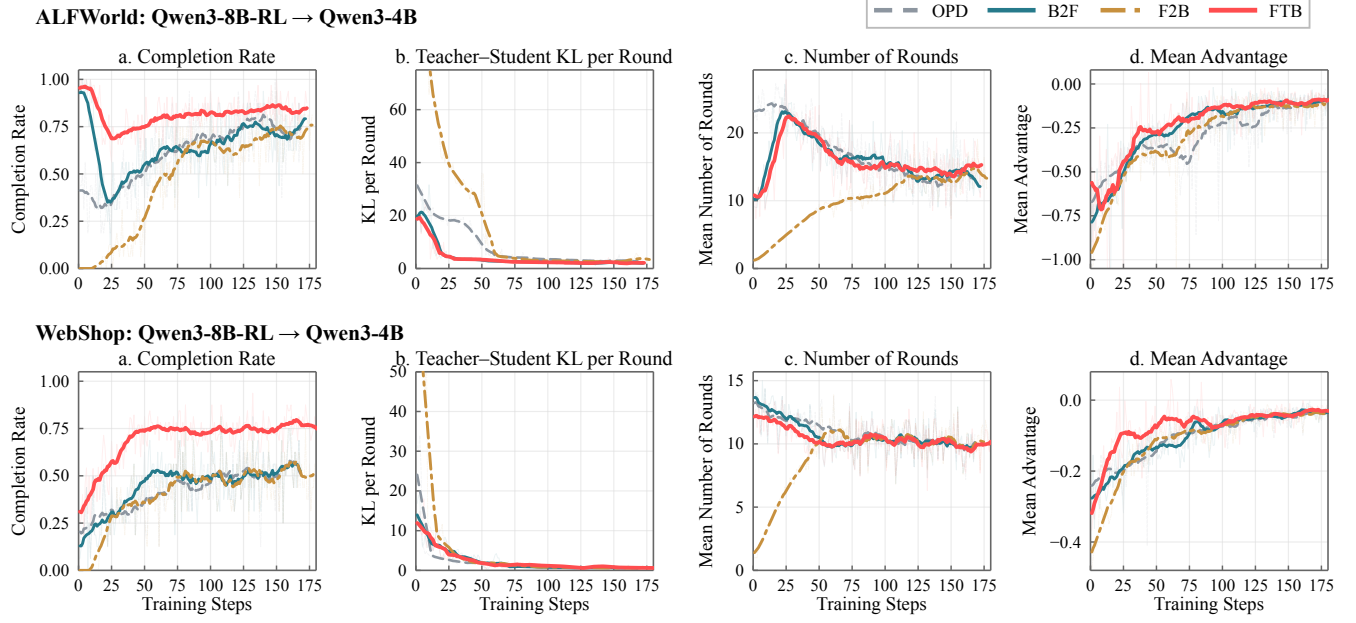


Figure 3: Training dynamics on ALFWorld and WebShop under the Qwen3-8B-RL teacher $\rightarrow$ Qwen3-4B student setting. The top and bottom rows correspond to ALFWorld and WebShop, respectively. From left to right, the four columns report the trajectory completion rate, teacher-student KL divergence per interaction round, mean number of interaction rounds, and mean sampled distillation advantage over training.

Method	ALFWorld		WebShop	
	SR $\uparrow$	Score $\uparrow$	SR $\uparrow$	
Qwen3-8B-RL teacher $\rightarrow$ Qwen3-4B student				
Student (zero-shot)	20.9	32.9	29.0	
Teacher (zero-shot)	79.9	62.6	61.0	
OPD	67.4 $\pm$ 3.7	63.7 $\pm$ 2.5	64.0 $\pm$ 1.6	
TCOD-F2B	66.5 $\pm$ 1.3	63.3 $\pm$ 2.6	64.7 $\pm$ 2.1	
TCOD-B2F	68.6 $\pm$ 2.3	61.7 $\pm$ 1.7	63.3 $\pm$ 1.2	
Guided-OPD	72.6 $\pm$ 3.4	63.8 $\pm$ 3.9	62.0 $\pm$ 5.0	
FTB w/o Bridge Exec.	70.8 $\pm$ 1.6	63.0 $\pm$ 0.8	63.0 $\pm$ 1.4	
FTB (Ours)	74.1 $\pm$ 3.1	65.7 $\pm$ 1.2	66.3 $\pm$ 0.5	

Table 2: Results with an RL-trained teacher, reported as mean $\pm$ standard deviation over three seeds; **blue** and **red** backgrounds denote the best and second-best trained methods for each reported metric, respectively.

Training Dynamics. Figure 3 compares the training dynamics under the Qwen3-8B-RL teacher $\rightarrow$ Qwen3-4B student setting. On both benchmarks, FTB improves the completion rate earlier and rapidly reduces the teacher-student KL difference. As training progresses, the numbers of interaction rounds gradually converge to similar levels across methods, while FTB achieves a faster increase in mean distillation advantage, indicating more effective optimization of the student policy during training.

4.3 Ablation Study

Following the Qwen3-8B-RL teacher $\rightarrow$ Qwen3-4B student setting in Table 2, we conduct the ablation study in Table 3 to examine candidate position selection, teacher bridge execution, and future trajectory validation. *FTB w/ Random*

Method	ALFWorld		WebShop	
	SR $\uparrow$	Score $\uparrow$	SR $\uparrow$	
Qwen3-8B-RL teacher $\rightarrow$ Qwen3-4B student				
Student (zero-shot)	20.9	32.9	29.0	
Teacher (zero-shot)	79.9	62.6	61.0	
TCOD-B2F	68.6 $\pm$ 2.3	61.7 $\pm$ 1.7	63.3 $\pm$ 1.2	
FTB w/ Random Turn	62.3 $\pm$ 3.0	58.3 $\pm$ 1.3	60.3 $\pm$ 1.2	
FTB w/o Bridge Exec.	70.8 $\pm$ 1.6	63.0 $\pm$ 0.8	63.0 $\pm$ 1.4	
FTB w/o Future Validation	71.9 $\pm$ 5.8	65.0 $\pm$ 1.1	64.3 $\pm$ 3.2	
FTB (Full)	74.1 $\pm$ 3.1	65.7 $\pm$ 1.2	66.3 $\pm$ 0.5	

Table 3: Ablation results with an RL-trained teacher, reported as mean $\pm$ standard deviation over three seeds; **blue** and **red** backgrounds denote the best and second-best trained methods for each reported metric, respectively.

Turn replaces the turn with the largest teacher-student disagreement with a randomly selected turn, evaluating the importance of locating critical intervention positions. *FTB w/o Bridge Exec.* does not execute the teacher bridge in the environment. It determines whether to include the corresponding teacher action in training by comparing the teacher preferred token ratios of the trajectory segments before and after the candidate position in the original student trajectory. *FTB w/o Future Validation* executes the teacher bridge and lets the student continue from the resulting state, but no longer filters the teacher action according to whether the subsequent trajectory achieves a higher teacher preferred token ratio than the original trajectory. Random turn selection causes the largest performance degradation, while removing bridge execution

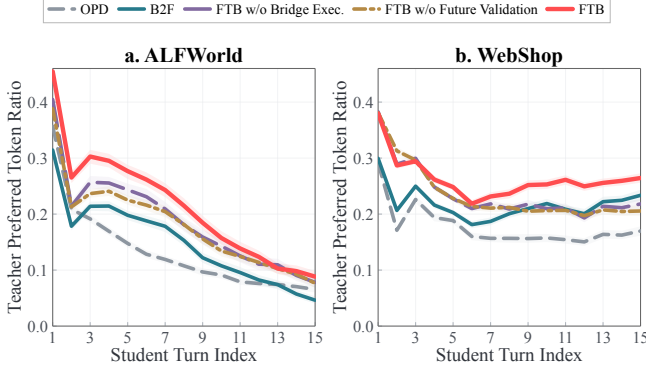


Figure 4: Teacher preferred token ratio over student turns on ALFWorld and WebShop.

or future validation also consistently reduces performance. The full FTB achieves the best results across all metrics.

5 Analysis

Teacher Preferred Token Ratio over Student Turns.

We measure the teacher preferred token ratio at each turn after the student takes control. As shown in Figure 4, FTB achieves a higher teacher preferred token ratio over most turns on both ALFWorld and WebShop, whereas removing bridge execution or future validation generally degrades it. Moreover, although bridge tokens constitute only 8.8% of the tokens used for optimization, they contribute 75.2% of the total positive distillation advantage. These results suggest that future trajectory validation identifies a small set of informative teacher actions that improve the alignment of subsequent student trajectories with the teacher.

Bridge Characteristics. We analyze retained teacher bridges in terms of their trigger positions, disagreement dynamics, and modifications to student actions.

Trigger Positions and Acceptance Rates. Figure 5 (a) shows the cumulative distribution of retained bridges over student turns and the fraction of candidates with high disagreement accepted by the future validation gate. More than half of the bridges occur at the first student turn, and approximately 85% occur within the first five turns. The acceptance rate also decreases over turns, from 39.7% at the first turn to 18.6% at the second and approximately 4–7% at later turns. Early positions are therefore selected more often and are more likely to yield improved student continuations. This is consistent with our earlier observation that early deviations have stronger effects on subsequent interactions.

Disagreement at Trigger Positions. Figure 5(b) shows the KL divergence between the teacher and student at the selected trigger positions over training. Under full FTB, the KL divergence decreases rapidly and remains low. Removing bridge execution or future validation causes it to rebound during later training. Local disagreement can identify positions that may benefit from correction, but cannot determine by itself whether a teacher action is useful. Bridge execution and future validation further

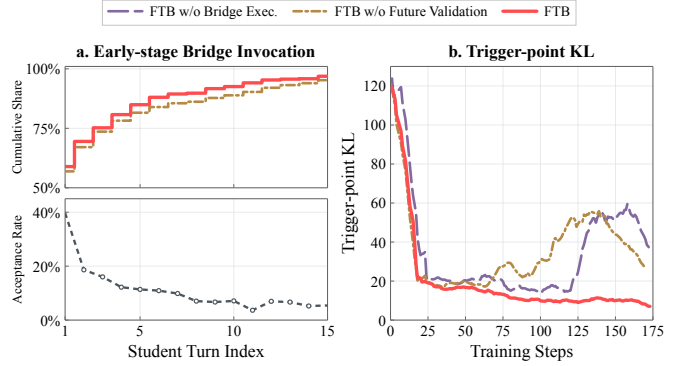


Figure 5: Bridge invocation, acceptance, and trigger point disagreement during training.

Method	Qwen3-32B teacher	
	Qwen3-1.7B student	Qwen3-4B student
OPD	481.5 (1.00 $\times$)	908.2 (1.00 $\times$)
TCOD-B2F	296.3 (0.62 $\times$)	500.8 (0.55 $\times$)
TCOD-F2B	193.1 (0.40 $\times$)	212.2 (0.23 $\times$)
FTB (Ours)	416.7 (0.86$\times$)	577.9 (0.64$\times$)

Table 4: Training time per step on ALFWorld, averaged over 200 steps under the same $8\times A100$ setup. Relative costs with respect to OPD are shown in parentheses.

remove candidates that fail to improve the subsequent trajectory.

Modifications to Student Actions. Retained bridges are rarely identical to the original student actions. In ALFWorld, 91.5% of bridges modify the action. Among them, 58.9% preserve the action type but change its target or argument, while 32.7% change the action type. In WebShop, 68.9% of bridges modify the action. Among them, 64.2% adjust the arguments while preserving the search or click type, and only 4.8% change the action type. These patterns show that retained bridges can either alter the action structure or preserve the high level operation while making fine grained corrections to the student policy.

Training Efficiency. Table 4 reports the average runtime per step on ALFWorld. Despite performing an additional teacher bridge and a short student validation continuation, FTB is consistently faster than OPD across student scales, achieving 0.86 $\times$ and 0.64 $\times$ of OPD’s runtime for the 1.7B and 4B students, respectively. The reason is that OPD requires a full student rollout at every training step and thus scales strongly with student size, whereas FTB inherits the shortened student rollouts of the B2F curriculum and only adds bounded validation overhead on top. Compared with B2F, FTB introduces roughly 120s and 77s of additional runtime on the 1.7B and 4B students, respectively, indicating that the validation overhead remains moderate and does not grow proportionally with student size. Overall, future validation adds no substantial training-time penalty, and its efficiency advantage over OPD grows with student size.

6 Conclusion

We propose FutureBridge-OPD (FTB) to mitigate error accumulation in multi-turn agentic on-policy distillation by identifying high-disagreement states, executing local teacher bridges, and validating them through induced student continuations. Unlike methods that assess guidance using local teacher-student disagreement, FTB explicitly evaluates whether an intervention redirects future student behavior toward regions with stronger teacher preference. Experiments on ALFWorld, WebShop, and ScienceWorld demonstrate consistent gains across student scales and teacher configurations. Ablations and trajectory analyses further confirm the complementary roles of candidate localization, bridge execution, and future validation. These results establish future trajectory validation as an effective way to select useful guidance in agentic distillation.

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A Overview

This supplement reports the intervention analysis motivating FTB, experimental settings and additional implementation details, formal results for sampled-token reverse-KL estimation and future-trajectory validation, sensitivity to the validation horizon, distinctions from closely related methods, and limitations.

B Motivating Intervention Analysis

We analyze 1,000 trajectories that are unsuccessful according to the terminal task-success criterion of each benchmark, comprising 500 ALFWorld, 250 WebShop, and 250 ScienceWorld trajectories.

For each trajectory, we identify the eligible Student-controlled turn with the largest Teacher-Student disagreement, replace the original Student response at that turn with a Teacher response, and then let the same frozen Student policy complete the remaining interaction.

Because the available outcome signals differ across benchmarks, we use a benchmark-specific comparison rule. For ALFWorld, only binary terminal task success is available for this analysis. An intervention is therefore classified as improved if it converts the originally failed episode into a successful one and unchanged otherwise; a degraded outcome is not applicable under this binary comparison. For WebShop and ScienceWorld, which additionally provide graded process rewards, we compare the benchmark-specific trajectory-level process score after intervention with that of the original trajectory and classify the intervention as improved, unchanged, or degraded.

For the random baseline, the intervention position is sampled uniformly from the same set of eligible Student-controlled turns and evaluated using the same benchmark-specific comparison rule.

As shown in Table 5, across all 1,000 trajectories, interventions at high-disagreement positions improve 309 trajectories (30.9%), compared with 57 trajectories (5.7%) when the intervention position is selected at random. Among the 500 WebShop and ScienceWorld trajectories, for which decreases are measurable using a graded process score, 95 high-disagreement interventions (19.0%) degrade trajectory performance.

These results suggest that high-disagreement positions are more likely to contain critical errors, while local disagreement alone is insufficient to determine whether a Teacher intervention is beneficial.

Teacher preference after local intervention. We further examine whether a local Teacher intervention changes the density of positive distillation signals in the subsequent Student trajectory. Following the Teacher-preferred token ratio defined in the main paper, we first compute the fraction of Student-generated continuation tokens for which the Teacher assigns a higher likelihood than the frozen Student, and then average this ratio across trajectories.

Before intervention, the mean Teacher-preferred token ratio is 25.45%. After replacing the Student response at the selected high-disagreement position with a Teacher response and returning control to the same frozen Student, the ratio increases to 28.20%. This corresponds to an absolute increase of 2.75 percentage points and a relative increase of 10.8%, computed from the unrounded values.

This analysis is conducted over all high-disagreement interventions before applying the future-preference gate. Therefore, the observed increase is not a direct consequence of selecting trajectories according to the same Teacher-preference criterion. The result suggests that local Teacher guidance can shift the subsequent Student continuation toward regions with denser positive distillation signals.

Training Signal and Evaluation Protocol. FTB does not use environment rewards as online optimization signals or in candidate localization and future-trajectory validation. Following TCOB-B2F, however, the temporal curriculum is initialized from pre-collected successful reference trajectories. Thus, beyond the success-based trajectory selection inherited from B2F, FTB uses distillation signals derived from the Teacher and Student policies during interactive training.

Candidate localization uses sampled Teacher-Student disagreement, and bridge retention is determined by the Teacher-preferred token ratio along paired Student continuations. Environment outcomes, including the benchmark-specific signals used in the motivating intervention analysis above, are used only for downstream performance evaluation and post-hoc analysis. They are not incorporated as additional reward signals into the FTB-specific candidate localization, bridge retention, or optimization procedure.

This protocol allows the comparison with TCOB-B2F to isolate the effect of future-trajectory validation: FTB introduces no additional reward-based training supervision beyond the successful reference trajectories already used by the underlying B2F curriculum.

In FTB, the Teacher-preferred token ratio measures the density of positive distillation signals along the subsequent Student continuation. It is used to select Teacher guidance that is consistent with the OPD objective, and its effectiveness is evaluated through the downstream performance of the resulting Student policy.

C Experimental Details

This section provides the benchmark splits, model configurations, generation parameters, optimization settings, and implementation details used in our experiments. Within each benchmark and Teacher-Student configuration, all methods use the same Student initialization, Teacher checkpoint, training steps, and evaluation setup.

C.1 Benchmarks and Evaluation

We evaluate all methods on ALFWorld, WebShop, and ScienceWorld. The data splits, interaction limits, and evaluation metrics are summarized in Table 6.

Benchmark	# Trajectories	High-disagreement Improve	No Change	Degrade	Random Improve
ALFWorld	500	159 (31.8%)	341 (68.2%)	N/A	28 (5.6%)
WebShop	250	92 (36.8%)	119 (47.6%)	39 (15.6%)	18 (7.2%)
ScienceWorld	250	58 (23.2%)	136 (54.4%)	56 (22.4%)	11 (4.4%)
<i>Aggregated statistics</i>					
All benchmarks	1,000	309 (30.9%)	—	—	57 (5.7%)
Graded-score benchmarks	500	150 (30.0%)	255 (51.0%)	95 (19.0%)	29 (5.8%)

Table 5: Teacher intervention results on trajectories that fail the terminal task-success criterion. ALFWorld uses binary terminal success, whereas WebShop and ScienceWorld additionally use their graded trajectory-level process scores.

Benchmark	Training Data	Evaluation Data	Max. Steps	Metrics
ALFWorld	Standard training split	Unseen evaluation split	30	Success rate
WebShop	Subset of the training split (aligned with TCOB)	100 held-out sessions	15	Task score and success rate
ScienceWorld	Task-type training split	Disjoint task-type split	30	Task score and success rate

Table 6: Benchmark data, maximum numbers of environment interactions, and evaluation metrics.

For ALFWorld, success rate is the percentage of episodes completed successfully. WebShop reports task score and success rate. Let $r_i \in [0, 1]$ denote the environment reward for session i . The two metrics are computed as

$$\text{Score} = \frac{100}{N} \sum_{i=1}^N r_i, \quad \text{SR} = \frac{100}{N} \sum_{i=1}^N \mathbf{1}[r_i > 0.5]. \quad (11)$$

The zero-shot and offline WebShop evaluations use 100 sessions with indices from 4096 to 4195. For ScienceWorld, we report task score and success rate.

C.2 Models and Training Configuration

The main experiments use Qwen3-32B as the Teacher and Qwen3-1.7B or Qwen3-4B as the Student. All methods within the same experimental setting share the same frozen Teacher checkpoint and Student initialization.

The experiments with an RL-trained Teacher use Qwen3-8B as the Teacher and Qwen3-4B as the Student. The Teacher checkpoint and Student initialization are fixed across methods in this setting.

Generation and optimization settings. All generation processes during training use a temperature of 1.0. This setting applies to Student rollouts, Teacher prefixes, and Teacher bridge actions. Student rollouts retain token-level log-probabilities for OPD supervision, Teacher–Student disagreement, and continuation alignment. We do not use beam search, multi-candidate search, or candidate reranking.

Evaluation uses a temperature of 0.4, with thinking mode disabled for all main results.

We set the distillation coefficient to $\beta = 1.0$. Both B2F and accepted bridge samples are optimized using the same PPO policy loss. During PPO optimization, the importance ratio is restricted to $[0.8, 1.2]$. For negative advantages, the auxiliary lower-bound factor is set to 3.0. Losses are averaged over valid response tokens, and the gradient norm is capped at 1.0.

The shared training hyperparameters are listed in Table 7.

Hyperparameter	Value
Optimizer	AdamW
Learning rate	1×10^{-6}
Distillation coefficient β	1.0
Importance-ratio clipping interval	$[0.8, 1.2]$
Negative-advantage lower-bound factor	3.0
Loss aggregation	Token average
Rollout batch size	16
Train batch size	64
Gradient clipping	1.0
Training steps	200
Maximum prompt length	10,240 tokens
Maximum response length	512 tokens

Table 7: Shared generation and optimization settings.

C.3 RL Teacher Training

We additionally train Qwen3-8B agents with GiGPO to construct stronger Teachers for the subsequent distillation experiments. Their shared and benchmark-specific settings are summarized together in Table 8.

The maximum numbers of environment interactions are 50 and 15 for ALFWorld and WebShop, respectively. Validation rollouts use sampling with a temperature of 0.4, and thinking mode is disabled during RL training.

C.4 Additional Implementation Details

Beyond the procedure described in the main paper, the B2F prefix is shortened by one environment step every five optimization steps. Each Student trajectory contains at most one bridge attempt, and the final Student-controlled turn is excluded because it has no subsequent continuation for validation.

At the selected position, the Teacher generates one bridge candidate with a temperature of 1.0. The same frozen Student is then rolled out for $H = 3$ Student-controlled turns on both the original and bridge branches. Before this paired validation, the environment is reinitialized and all interactions preceding the selected position are replayed to recover the same state.

Hyperparameter	ALFWorld RL Teacher	WebShop RL Teacher
Base model	Qwen3-8B	Qwen3-8B
RL algorithm	GiGPO	GiGPO
Optimizer	AdamW	AdamW
Learning rate	1×10^{-6}	1×10^{-6}
Weight decay	0.01	0.01
Discount factor γ	0.95	0.95
Rollout group size	8	8
Invalid-action penalty	0.1	0.1
Maximum prompt length	4,096 tokens	4,096 tokens
Maximum response length	512 tokens	512 tokens
Maximum environment interactions	50	15
Train batch size	32	16
PPO mini-batch size	64	64
PPO micro-batch size per GPU	4	4
Rollout micro-batch size per GPU	4	4
Log-prob micro-batch size per GPU	8	8
Tensor parallel size	2	2
Number of GPUs	8	8

Table 8: Configurations used to train the Qwen3-8B RL Teachers for ALFWorld and WebShop.

The original B2F samples and accepted bridge samples are optimized through separate loss terms. Specifically, the overall FTB objective is

$$\mathcal{L}_{\text{FTB}} = \mathcal{L}_{\text{B2F}} + \lambda_{\text{br}} \mathcal{L}_{\text{bridge}}, \quad \lambda_{\text{br}} = 1. \quad (12)$$

The B2F loss and bridge loss are independently averaged over their valid response tokens and then summed with equal coefficients. Bridge supervision therefore augments rather than replaces the standard B2F objective, and both sample types use the same optimization settings.

The benchmark-specific prompt templates follow TCOB and are provided with the released code.

D Formal Analysis

This section complements the definitions in the main paper. We first clarify the sampled-token reverse-KL estimator, then formalize how the paired continuation difference estimates the underlying future-preference advantage, and give a local interpretation of accepted bridge optimization.

D.1 Sampled-Token Estimation of Reverse KL

The main paper defines candidate disagreement using tokens sampled by the Student. For any visited token context c ,

$$\mathbb{E}_{x \sim \pi_{\bar{\theta}}(\cdot | c)} \left[\log \frac{\pi_{\bar{\theta}}(x | c)}{\pi_{\phi}(x | c)} \right] = D_{\text{KL}}(\pi_{\bar{\theta}}(\cdot | c) \| \pi_{\phi}(\cdot | c)). \quad (13)$$

Recent OPD methods commonly estimate the reverse-KL objective through Student-sampled tokens rather than explicitly summing over the full vocabulary (Zhong et al., 2026; Xing et al., 2026). FTB follows the same sampled-token estimator. Its realized value is used only to rank candidate positions; bridge utility is decided by the paired future validation.

D.2 Estimating the Future-Preference Advantage

Fix the intervention context

$$\mathcal{C} = (h_{t^*}, a^{\text{br}}, a^{\text{stu}}). \quad (14)$$

Let

$$X^{\text{br}} = \rho(\xi^{\text{br}}), \quad X^{\text{base}} = \rho(\xi^{\text{base}}), \quad (15)$$

where both variables lie in $[0, 1]$. The true auxiliary future-preference advantage of the bridge is

$$\Delta_{\phi, \bar{\theta}}^H = \mathbb{E}[X^{\text{br}} - X^{\text{base}} | \mathcal{C}]. \quad (16)$$

The gate uses the one-pair estimate

$$\hat{\Delta}_1^H = X^{\text{br}} - X^{\text{base}}. \quad (17)$$

Proposition 1 (Unbiased paired estimate). If the two branches have the intended marginal rollout distributions, then

$$\mathbb{E}[\hat{\Delta}_1^H | \mathcal{C}] = \Delta_{\phi, \bar{\theta}}^H. \quad (18)$$

Proof. By linearity of conditional expectation,

$$\begin{aligned} \mathbb{E}[\hat{\Delta}_1^H | \mathcal{C}] &= \mathbb{E}[X^{\text{br}} | \mathcal{C}] - \mathbb{E}[X^{\text{base}} | \mathcal{C}] \\ &= \Delta_{\phi, \bar{\theta}}^H. \end{aligned} \quad (19)$$

□

The proposition does not require the two branches within a pair to be independent; it only requires each branch to have the intended marginal distribution.

To make the relation to the true value more explicit, consider K independent continuation pairs and define

$$\hat{\Delta}_K^H = \frac{1}{K} \sum_{k=1}^K (X_k^{\text{br}} - X_k^{\text{base}}). \quad (20)$$

Proposition 2 (Concentration around the true advantage). For every $\epsilon > 0$,

$$\Pr\left(\left|\hat{\Delta}_K^H - \Delta_{\phi, \bar{\theta}}^H\right| \geq \epsilon \mid \mathcal{C}\right) \leq 2 \exp\left(-\frac{K\epsilon^2}{2}\right). \quad (21)$$

Proof. For each pair, $Z_k = X_k^{\text{br}} - X_k^{\text{base}} \in [-1, 1]$ and $\mathbb{E}[Z_k \mid \mathcal{C}] = \Delta_{\phi, \bar{\theta}}^H$. Applying Hoeffding’s inequality to the independent bounded variables $\{Z_k\}_{k=1}^K$ gives Eq. (21). $\square$

A useful one-sided consequence is that, for any margin $m > 0$ and any fixed context $\mathcal{C}$ satisfying $\Delta_{\phi, \bar{\theta}}^H \leq 0$,

$$\Pr\left(\hat{\Delta}_K^H \geq m \mid \mathcal{C}\right) \leq \exp\left(-\frac{Km^2}{2}\right). \quad (22)$$

Thus, a larger positive observed bridge gain is exponentially less likely to arise from a bridge whose true future-preference advantage is non-positive. With more paired samples, the estimate approaches the true auxiliary future-preference advantage. FTB uses $K = 1$ for efficiency, so the gate remains a noisy estimate.

D.3 Local Interpretation of Bridge Optimization

The accepted bridge samples are optimized together with the B2F samples using the PPO policy loss. Let

$$\ell_i(\theta) = \log \pi_\theta(x_i^{\text{br}} \mid c_i^{\text{br}}), \quad \ell_i^\phi = \log \pi_\phi(x_i^{\text{br}} \mid c_i^{\text{br}}), \quad (23)$$

$$\delta_i(\theta) = \ell_i(\theta) - \ell_i^\phi. \quad (24)$$

Consider the accepted-token discrepancy

$$\mathcal{D}_{\text{bridge}}(\theta) = \mathbb{E}\left[\frac{\beta g(\tau)}{2M_{\text{br}}} \sum_{i=1}^{M_{\text{br}}} \delta_i(\theta)^2\right]. \quad (25)$$

Proposition 3 (First-order bridge update). Treat the sampled bridge data, advantages, and gate as fixed. At the rollout parameters $\theta = \bar{\theta}$, the gradient of the clipped PPO loss on accepted bridge tokens equals the gradient of $\mathcal{D}_{\text{bridge}}$:

$$\nabla_\theta \mathcal{L}_{\text{PPO}}^{\text{bridge}}(\theta) \Big|_{\bar{\theta}} = \nabla_\theta \mathcal{D}_{\text{bridge}}(\theta) \Big|_{\bar{\theta}}. \quad (26)$$

Proof. Let

$$r_i(\theta) = \frac{\pi_\theta(x_i^{\text{br}} \mid c_i^{\text{br}})}{\pi_{\bar{\theta}}(x_i^{\text{br}} \mid c_i^{\text{br}})}. \quad (27)$$

At $\theta = \bar{\theta}$, $r_i = 1$ and all ratio-clipping branches agree locally, while

$$\nabla_\theta r_i(\bar{\theta}) = \nabla_\theta \ell_i(\bar{\theta}). \quad (28)$$

Moreover,

$$A_i^{\text{br}} = \beta \left(\ell_i^\phi - \ell_i(\bar{\theta}) \right) = -\beta \delta_i(\bar{\theta}). \quad (29)$$

Therefore, the bridge part of the PPO gradient at the rollout policy is

$$\begin{aligned} \nabla_\theta \mathcal{L}_{\text{PPO}}^{\text{bridge}}(\theta) \Big|_{\bar{\theta}} &= -\mathbb{E} \left[\frac{g(\tau)}{M_{\text{br}}} \sum_i A_i^{\text{br}} \nabla_\theta r_i(\bar{\theta}) \right] \\ &= \mathbb{E} \left[\frac{\beta g(\tau)}{M_{\text{br}}} \sum_i \delta_i(\bar{\theta}) \nabla_\theta \ell_i(\bar{\theta}) \right], \end{aligned} \quad (30)$$

which is exactly $\nabla_\theta \mathcal{D}_{\text{bridge}}(\bar{\theta})$. $\square$

For a sufficiently small gradient step from $\bar{\theta}$, a first-order expansion therefore gives

$$\mathcal{D}_{\text{bridge}}(\theta^+) = \mathcal{D}_{\text{bridge}}(\bar{\theta}) - \eta \|\nabla_\theta \mathcal{D}_{\text{bridge}}(\bar{\theta})\|_2^2 + O(\eta^2). \quad (31)$$

Hence, the accepted bridge update locally reduces the sampled Teacher–Student log-probability discrepancy on the bridge tokens. This is a local optimization statement and does not imply monotonic improvement in environment return or global policy divergence.

Taken together, Propositions 1–2 formalize how the observed bridge gain estimates the true auxiliary future-preference advantage, while Proposition 3 explains how an accepted bridge locally moves the Student toward the Teacher on the selected bridge tokens.

E Sensitivity to the Validation Horizon

Beyond the shared training and generation settings, the only method-specific tunable hyperparameter introduced by FTB is the validation horizon H , which controls the number of Student-controlled continuation turns used to evaluate a candidate Teacher bridge. We use $H = 3$ in the main experiments and further evaluate $H \in \{1, 3, 5\}$ under a fixed 200-step optimization schedule. These sensitivity results are obtained from single-seed runs under the main Qwen3-32B Teacher to Qwen3-1.7B Student configuration.

As shown in Table 9, FTB remains effective across all tested horizons. The best value of H varies across benchmarks and evaluation metrics: $H = 5$ achieves the highest success rates on ALFWorld and WebShop, whereas $H = 3$ obtains the highest WebShop score and ScienceWorld success rate. On ScienceWorld, $H = 5$ instead achieves the highest task score. The absence of a consistently dominant horizon, together with the broadly comparable performance across the tested values, indicates that FTB is not strongly sensitive to the exact choice of H .

F Comparison with Closely Related Work

We focus on methods whose mechanisms are most likely to be confused with FTB. FTB executes the original Student action and a candidate Teacher bridge from the same restored environment state. The same frozen Student then generates both bounded continuations, which are used only to decide whether the Teacher bridge is retained.

Guided-OPD. Guided-OPD mixes Teacher- and Student-generated turns within each rollout and decreases the Teacher-intervention probability through a curriculum (Li

Horizon H	ALFWorld	WebShop		ScienceWorld	
	SR $\uparrow$	SR $\uparrow$	Score $\uparrow$	SR $\uparrow$	Score $\uparrow$
1	39.0	53.0	55.4	3.5	27.3
3	41.0	60.0	60.2	7.5	26.0
5	43.0	62.0	56.7	4.5	28.9

Table 9: Single-seed sensitivity to the future-validation horizon H at 200 optimization steps under the main Qwen3-32B Teacher to Qwen3-1.7B Student configuration. Success rates and task scores are reported on a 0–100 scale. The best horizon varies across benchmarks and metrics, indicating that FTB is not strongly sensitive to the exact choice of H .

et al., 2026a). A selected Teacher turn directly changes the rollout trajectory. FTB instead preserves the original Student branch as a comparison and validates each candidate Teacher action through the Student future it induces.

SAGE-OPD. SAGE-OPD uses environment feedback and Teacher judgment to assign turn-level intervention strengths, and further weights token-level OPD by Teacher confidence (Zhou et al., 2026a). It therefore changes how strongly an already executed Student response is distilled. FTB instead generates and executes an alternative Teacher action, then decides whether that action should be added as auxiliary supervision.

ReOPD. ReOPD reuses pre-collected Teacher trajectories as replayed prefixes and performs Student training without new environment interactions (Liao et al., 2026). Its main design problem is selecting a prefix distribution that balances Student relevance and Teacher reliability. FTB operates on states reached during interactive Student rollouts and uses environment restoration to compare the consequences of two executable actions.

TOPD. TOPD compares short Teacher and Student continuations from the same textual prefix, uses their near-future divergence to identify genuine reasoning forks, and injects trajectory discrepancy into a multi-token objective through trajectory alignment (Jiang and Ferraro, 2026). In FTB, both continuations are generated by the same frozen Student after different environment actions. The future signal is used as an acceptance gate rather than as a trajectory-alignment loss.

Relay-OPD. Relay-OPD detects a handoff when the Teacher prefers a reflection token while the Student’s top- K support contains no reflection token (Xu et al., 2026a). The Teacher then generates a bounded reasoning leg, after which the Student may resume, and the resulting relay trajectory is optimized. FTB does not directly trigger Teacher takeover from this local asymmetry; it retains a paired original branch and accepts the Teacher bridge only after comparing the two induced Student futures.

TCOD-B2F. TCOD-B2F progressively shortens a successful Teacher or expert prefix so that the Student takes control from increasingly earlier states (Wang et al., 2026a). FTB retains this temporal curriculum and adds action-level bridge selection, execution, and paired future validation inside the Student-controlled suffix.

G Limitations

Our experiments use Qwen3 teacher–student pairs with a shared tokenizer, and extending agentic distillation across model families with different tokenizers remains unvalidated. Computational constraints also limit our evaluation to a modest range of model scales, leaving larger teachers and students for future work.